

TASK : Body Measurement Extraction Research

Today, I tried to improve the body measurement accuracy to get better measurements, I can't say it's a full success because I couldn't implement to get the perfect body measures but it was better than what I implemented the previous day.

THE CODE:

```

1  import cv2
2  import mediapipe as mp
3  import time
4  import math
5
6
7  mp_pose = mp.solutions.pose
8  mp_drawing = mp.solutions.drawing_utils
9
10 pose = mp_pose.Pose(
11     static_image_mode=False,
12     model_complexity=2,
13     smooth_landmarks=True,
14     min_detection_confidence=0.7,
15     min_tracking_confidence=0.7
16 )
17
18 cap = cv2.VideoCapture(0)
19
20 cap.set(3, 1280)
21 cap.set(4, 720)
22
23 previous_time = 0
24 PIXEL_TO_CM = 0.15
25
26 while True:
27     success, frame = cap.read()
28     if not success:
29         break
30
31     frame = cv2.flip(frame, 1)
32
33     frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
34
35     rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
36
37     results = pose.process(rgb_frame)
38
39     if results.pose_landmarks:
40
41         mp_drawing.draw_landmarks(

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        mp_drawing.draw_landmarks(
            frame,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS,
            mp_drawing.DrawingSpec(
                color=(0, 255, 0),
                thickness=2,
                circle_radius=2
            ),
            mp_drawing.DrawingSpec(
                color=(255, 0, 0),
                thickness=2
            )
        )

        landmarks = results.pose_landmarks.landmark

        h, w, c = frame.shape
        left_shoulder = landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER.value]
        right_shoulder = landmarks[mp_pose.PoseLandmark.RIGHT_SHOULDER.value]
        left_hip = landmarks[mp_pose.PoseLandmark.LEFT_HIP.value]
        right_hip = landmarks[mp_pose.PoseLandmark.RIGHT_HIP.value]

        ls_x, ls_y = int(left_shoulder.x * w), int(left_shoulder.y * h)
        rs_x, rs_y = int(right_shoulder.x * w), int(right_shoulder.y * h)

        lh_x, lh_y = int(left_hip.x * w), int(left_hip.y * h)
        rh_x, rh_y = int(right_hip.x * w), int(right_hip.y * h)

        shoulder_distance_pixels = math.sqrt(
            (rs_x - ls_x) ** 2 + (rs_y - ls_y) ** 2
        )
        shoulder_width_cm = shoulder_distance_pixels * PIXEL_TO_CM
        hip_distance_pixels = math.sqrt(
            (rh_x - lh_x) ** 2 + (rh_y - lh_y) ** 2
        )
        hip_width_cm = hip_distance_pixels * PIXEL_TO_CM
        nose = landmarks[mp_pose.PoseLandmark.NOSE.value]
        left_ankle = landmarks[mp_pose.PoseLandmark.LEFT_ANKLE.value]

```

```

81
82     nx, ny = int(nose.x * w), int(nose.y * h)
83     ax, ay = int(left_ankle.x * w), int(left_ankle.y * h)
84
85     body_height_pixels = abs(ay - ny)
86
87     body_height_cm = body_height_pixels * PIXEL_TO_CM
88     cv2.line(frame, (ls_x, ls_y), (rs_x, rs_y), (255, 255, 0), 3)
89     cv2.line(frame, (lh_x, lh_y), (rh_x, rh_y), (0, 255, 255), 3)
90     cv2.putText(
91         frame,
92         f'Shoulder Width: {shoulder_width_cm:.1f} cm',
93         (20, 120),
94         cv2.FONT_HERSHEY_SIMPLEX,
95         0.7,
96         (0, 255, 255),
97         2
98     )
99     cv2.putText(
100         frame,
101         f'Hip Width: {hip_width_cm:.1f} cm',
102         (20, 160),
103         cv2.FONT_HERSHEY_SIMPLEX,
104         0.7,
105         (255, 255, 0),
106         2
107     )
108     cv2.putText(
109         frame,
110         f'Estimated Height: {body_height_cm:.1f} cm',
111         (20, 200),
112         cv2.FONT_HERSHEY_SIMPLEX,
113         0.7,
114         (0, 255, 0),
115         2
116     )
117     current_time = time.time()
118     fps = 1 / (current_time - previous_time)
119     previous_time = current_time
120     cv2.putText(
121         frame,
122         f'FPS: {int(fps)}',

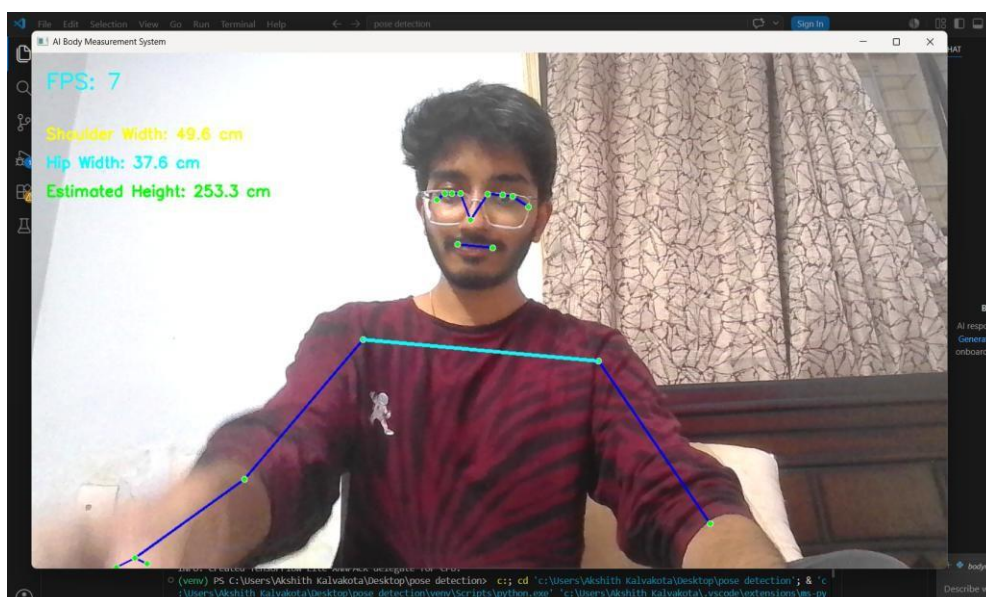
```

```

122         frame,
123         f'FPS: {int(fps)}',
124         (20, 50),
125         cv2.FONT_HERSHEY_SIMPLEX,
126         1,
127         (255, 255, 0),
128         2
129     )
130     # Show Frame
131     cv2.imshow("AI Body Measurement System", frame)
132     if cv2.waitKey(1) & 0xFF == ord('q'):
133         break
134     cap.release()
135     cv2.destroyAllWindows()

```

OUTPUT:



The accuracy of body measurement extraction was analyzed by testing the model on different images with varying poses, clothing styles, and image quality. It was observed that the measurement results depended heavily on how clearly body landmarks were detected by the pose estimation model. Front-facing images with a straight posture and full body visibility produced more consistent and reliable results.

The model has to be trained with different objects, for example if we take a 15 cm ruler, We have to tell that its 15 cm so the model converts the pixels into that height and in future takes these references to give more accurate height estimation.

The measurements are not exactly correct because webcam distance changes pixel size. For more accurate real-world measurements, we need to be able to use:

- Depth sensors
- Calibration objects
- Multiple camera angles
- AI body scaling models

And for better body measurements, we should:

1. Stand straight facing the camera
2. Full body should be visible
3. Maintain proper distance
4. Have good lighting
5. Wear fitted clothes
6. Stand still for 3-4 seconds